



Research Net

Accelerate research and product development with an agentic automation platform for the complete product lifecycle

<https://www.researchnet.ai/>

AI Alliance: Research Net Series

April 2nd, 2026



The PROBLEM



Manual Analysis is
Slow and subjective



Professional Analytics
are Inaccessible

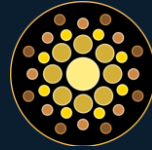


Feedback Lacks a
Visual Representation



Critical Match Insights
are Lost





Product Steps



Ultralytics and YOLO v8

Create the starting model

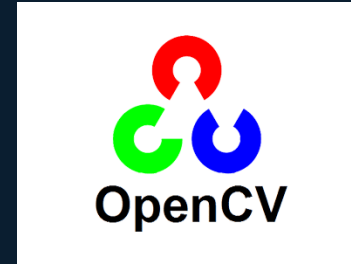
- most optimized tradeoff between accuracy and speed
- three main models: v8n, v8s, v8m
- works directly with COCO datasets right away
- can withstand live match analysis



Kaggle Datasets

Training the Model

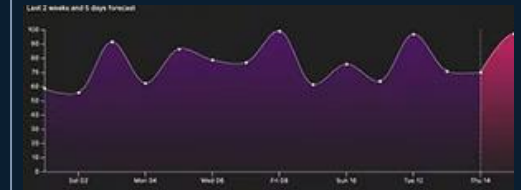
- the data set used right now has around 1800 images
- encompasses diverse match conditions
- captures authentic scenarios
- preparing models for motion blur, swing etc.



Live Video Integration

Using OpenCV to prepare model for live games

- OpenCV will help further automate the frame-by-frame output
- mapping video frames with field dimensions/player builds etc
- better trajectory/trail detection



Building the Product

Allowing players to track their stats post-match

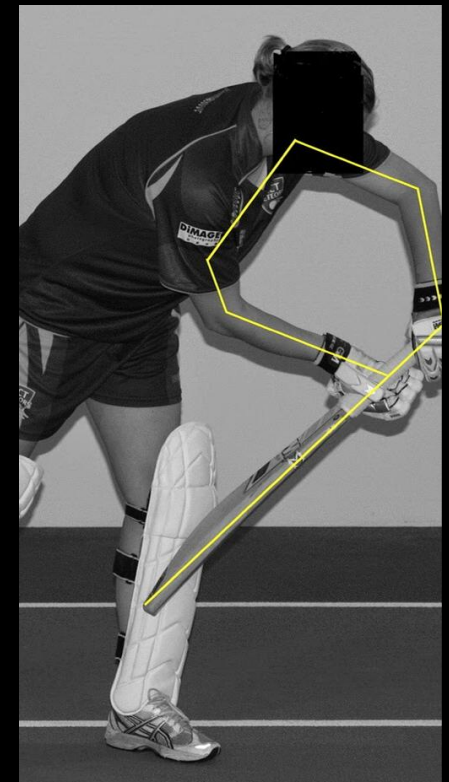
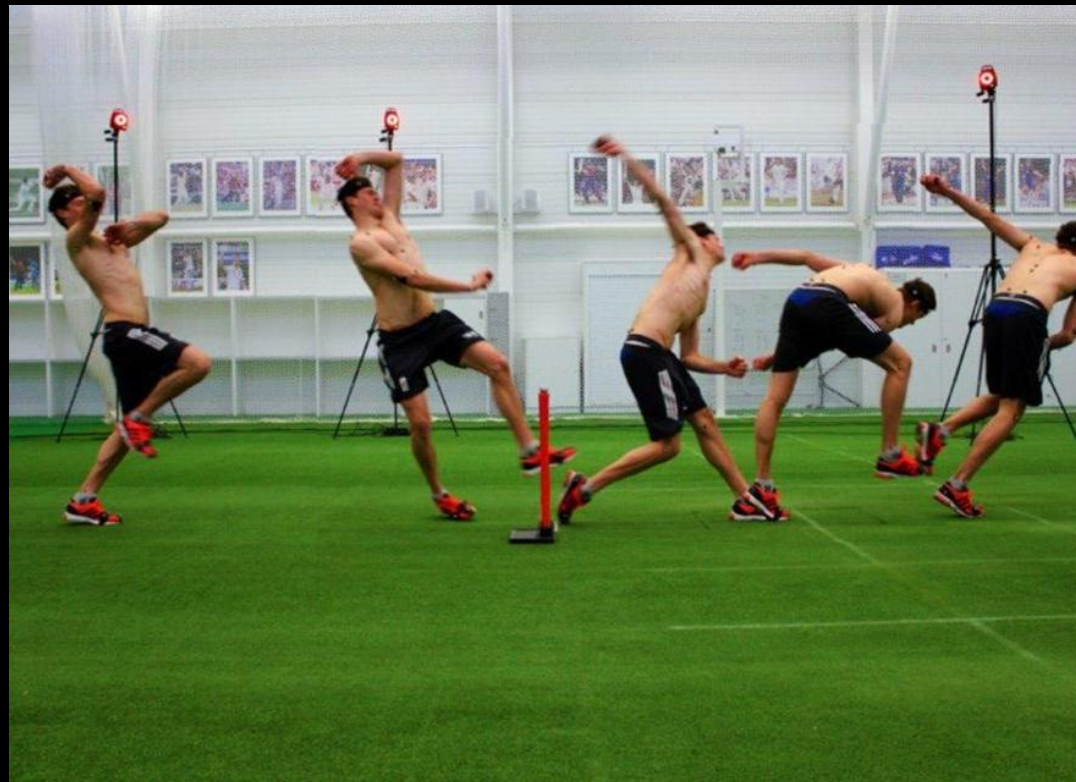
- intuitive GUI/dashboard for players and coaches to navigate easily
- advance metrics: more than just line and length for example
- every statistic should have the video it is referencing accessible/skeletal tracking



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Studying Biomechanics

- Quality fast bowling has four distinct phases: the run-up, the delivery stride, releasing the ball, and the follow-through, all of which should be optimized fully
- In the past few decades, coaches have already discovered optimization techniques such as knee bracing and upper trunk flexion to generate the most torque
- Different types of deliveries and their usefulness can also be detected by analyzing the biomechanics of fast bowler's wrist movement, and their grip of the seam on the ball
- It is also important to note that the most textbook biomechanical way might not be the most optimal for everyone but small tweaks can be made based on player preferences or game situation

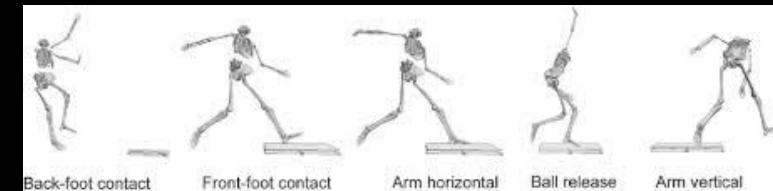
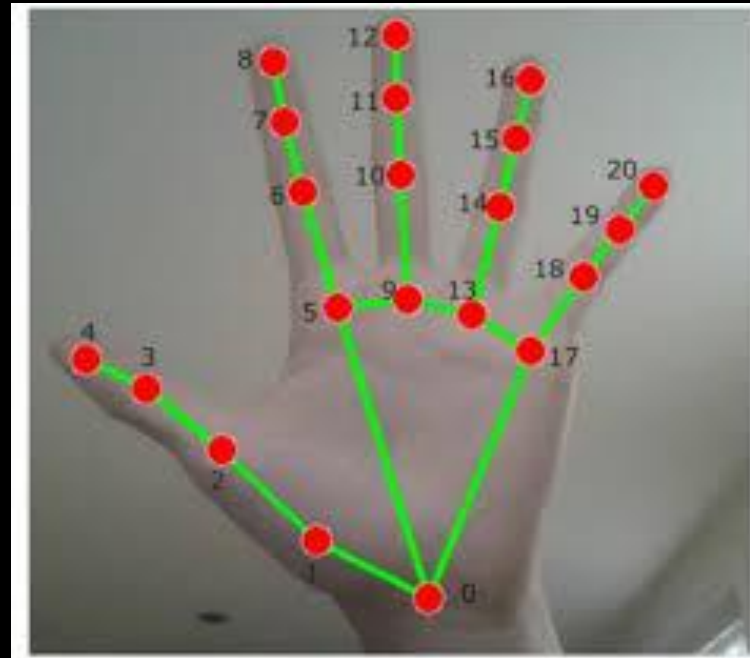
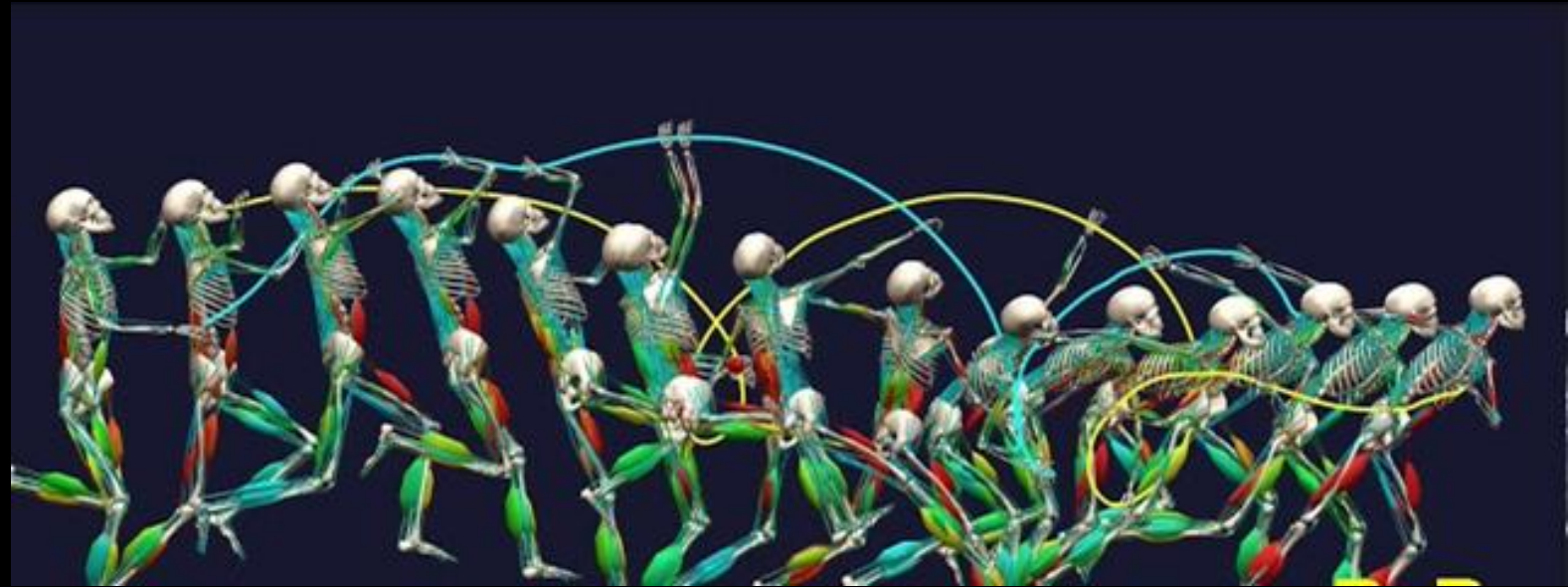




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Using Biomechanics to Analyze Skeletal Movements

- Libraries such as MediaPipe can extract around 20 joints (critical skeletal points as they call them) just within the hand. For one's entire body there can likely be close to 50 total points allowing the player and coach to analyze every specific muscle and joint movement
- The app will be designed to overlay these lines to gain real-time analytics and angles. This is also important because you want to ensure the app only recognizes movements that are possible for the human body.
- Dynamic Time Warping (DTW) algorithms shall be used in the product as well to align a player's video which they want to review into a professional database, allowing for accurate comparison, and insightful analytics.



The SOLUTION



Statistical Analysis
Rooted in Data



A Product Accessible
to even Club Players



A User-Friendly Easily-
Navigated Interface



Easy to Receive Post-
Match Insights





App Framework

- YOLOv8 successfully tracks the ball on the screen with a bounding box around it.
- MediaPipe identifies the 32 joints in the human body, will be used for in depth suggestions based on math and suggestions from esteemed cricket coaches
- Front-end allows for easy user experience when reviewing there games
- Videos can be imported from Youtube, PC files, or straight from your webcam
- The next steps for this app is as follows:
 - Learning how to retrieve the ball trajectory to track angle of shots
 - Calculating the point in which the ball bounces
 - Better role and player detection (face and jersey number)



Works Cited:

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Siddiqui, Hafeez Ur Rehman, et al. "Enhancing Cricket Performance Analysis with Human Pose Estimation and Machine Learning." *Sensors*, vol. 23, no. 15, art. 6839, 1 Aug. 2023, <https://doi.org/10.3390/s23156839>.

Thank You

Maake Asante Dhanyavadagalu Shukria
Vinaka Kaitos Maana Dankon Manana
감사합니다 Dank Je Dankscheen Kam Sah Hammida
Blagodaram Ngiyabonga Dziekuje Mauruuru Biyan
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Děkuji Nirringrazzjak Hvala Gracías Gratias Tibi
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Misaoatra Welalin Danke Kia Ora Kop Khun Khap Obrigado
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